

Limitations of Analytics on Anonymous Self-Reported Data

One limitation of analyzing anonymous self-reported data is that there is no way to verify whether all of the information is accurate. The GradCafe dataset is made up of submissions entered voluntarily by users, so the data may contain mistakes, missing fields, or inconsistent formatting. In addition, the people who choose to post their admissions results may not be representative of all graduate school applicants. Some applicants may be more likely to share their results than others, especially if they were accepted to a highly competitive program. Because of these factors, the results of the analysis should be viewed as trends within the GradCafe community rather than exact statistics for all graduate school applicants.

Some of the results were interesting and different from what I would expect based on national statistics. For example, the average GRE scores in the GradCafe dataset were noticeably higher than the averages reported for all GRE test takers. I think this is mainly due to selection bias. Many users on GradCafe are applying to competitive master's and PhD programs, particularly in STEM fields, where applicants often have stronger academic backgrounds and higher test scores. It is also possible that applicants with stronger profiles are more motivated to share their results online. As a result, the GradCafe dataset likely represents a subset of applicants who are more competitive than the average graduate school applicant. This assignment showed me that large datasets can provide useful insights, but it is important to understand where the data comes from and what biases may exist before drawing conclusions.